



REPORT

Understanding groundwater flow in a buried valley and bedrock aquifer system in Kurikka, Finland, using hydrochemical indicators and residence time tracers ^3H and ^{14}C

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Abstract

Growing demand for groundwater in western Finland has drawn attention to a regional network of buried valleys with high potential for domestic and municipal water supplies. This study aims at understanding the groundwater flow in the buried valley and bedrock aquifer system using hydrochemical data and residence time indicators (^3H , ^{14}C). A multivariate statistical analysis (hierarchical clustering on principal components) revealed six water groups reflecting the origins and different stages of groundwater evolution. Overall, a modern groundwater component (< 60 years old) and active flow system are present in all parts of the system. Longer residence times are found deeper within the buried valleys (^3H -free waters) and at greater depth in a few locations in the bedrock (^{14}C ages > 5000 years). The occurrence of geochemically evolved waters in the deeper, downgradient areas of buried valleys indicates suitable (in terms of quantity and quality) and safe (low vulnerability) water supply relative to demand. More broadly, the multi-tracer approach provides insights into the geochemical evolution of groundwater, revealing groundwater flow conditions, geological history and aquifer unit composition. Residence time indicators also illustrate the relative hydraulic isolation of buried valleys, which has implications on vulnerability and sustainable exploitation. To fully clarify the exchanges between bedrock and granular aquifers, a sampling scheme targeting deep flow paths beneath the Paloluoma and Kyrönjoki valleys would be necessary. This study will support the development of a numerical model of the aquifer system and contribute to informing future management decisions.

Keywords Hydrochemistry · Groundwater age · Conceptual models · Buried valley aquifer system · Multivariate statistical analysis

Introduction

Understanding the functioning of hydrosystems is a prerequisite for the sustainable management of groundwater resources. This is especially challenging in complex contexts, such as glacial buried valley aquifer systems where sediment heterogeneity, geological structures and exchanges

with bedrock influence the flow system and recharge processes. Nevertheless, buried valley aquifers can represent significant water resources (Cummings et al. 2012; Bajc et al. 2018; Steelman et al. 2018; Yager et al. 2019; Goetz et al. 2021; Kehler et al. 2024). However, specific challenges remain in understanding the groundwater flow dynamics, the hydraulic connections between bedrock and granular aquifers, and the groundwater residence time. At the regional scale, geochemical multi-tracer approaches are useful to understand groundwater origin, flow paths, and geochemical evolution. These approaches typically include hydrochemical data (major, minor, and trace elements), field parameters, isotopic tracers, and other indicators of groundwater residence time (Cloutier et al. 2006; Jaunat et al. 2012; Roques et al. 2014; Pétré et al. 2016; Beaudry et al. 2018). A multi-tracer approach is well suited to provide an understanding of regional flow systems because hydraulic data alone cannot provide clear indications about the provenance

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and evolution of groundwater at larger scale, especially in complex systems.

The large number of parameters usually included in a multi-tracer study represents a complex dataset and it is therefore necessary to apply adapted data analysis and interpretation tools, such as multivariate statistical analysis, to identify relationships, patterns, and processes affecting groundwater geochemistry (Cloutier et al. 2008; Voutsis et al. 2015; Yang et al. 2020; Liu et al. 2023). Besides geochemical composition, groundwater residence time indicators can further support the understanding of hydrosystems (Jaunat et al. 2012; Pétré et al. 2016; Beaudry et al. 2018). Groundwater dating tracers of young groundwater (less than 65 years) include chlorofluorocarbons (CFCs), sulfur hexafluoride (SF_6), tritium (^3H) and helium (^3He) while radiocarbon (^{14}C) and chloride-36 (^{36}Cl) are used to characterize old or very old groundwaters (Clark and Fritz 1997; Kazemi et al. 2006).

The goal of this study is to examine a regional network of buried valleys by characterizing both the granular aquifers within the buried valleys and the adjacent and underlying fractured bedrock. This buried valley aquifer system is expected to play a key role in the development of drinking water resources in western Finland, for the cities of Kurikka and Vaasa. Previous geological and seismic surveys suggested that the buried valley aquifer system might hold significant water resources, mainly due to the widespread occurrence of permeable sediments at depth within the buried valley (Malinowski et al. 2023; Putkinen et al. 2026). A major part of the study area lies within the *Suupohja passive ice area*, where minimal glacial erosion during the latest glaciation has resulted in the preservation of an ultra-thick (up to 120 m) sedimentary sequence.

Preliminary work by Purkamo et al. (2026) in the central Paloluoma Valley, indicated the presence of sub-modern water in the deepest aquifers of the buried valleys as well as in bedrock. However, this study also showed the significant limitations in using the $^3\text{H}/^3\text{He}$ method and CFCs/ SF_6 for groundwater dating in the Kurikka area. These limitations were due to contamination (CFCs), geological influence affecting SF_6 and radiogenic ^4He , and the absence of those tracers in several sampled locations. Rashid (2022) developed a numerical groundwater flow model of one of the buried valleys in the Kurikka area and simulations of long-term pumping tests (several months) indicated a substantial groundwater input from the bedrock to the sediments.

In this context, this study aimed to conduct a systematic hydrogeochemical assessment of the regional valley aquifer system to understand the hydrodynamics of the system, comprising both sediments and bedrock. In addition, the study aimed to determine the depth of active groundwater flow and the relationship between bedrock and sediment aquifers,

and the interactions between the different buried valleys. The study area includes four buried valleys connected to the main Kyrönjoki buried valley. Considering the limited applicability of some groundwater dating methods in this geological context (Purkamo et al. 2026) radiocarbon (^{14}C and $\delta^{13}\text{C}$ in dissolved inorganic carbon (DIC)) and ^3H were used as residence time indicators.

Materials and methods

Study area

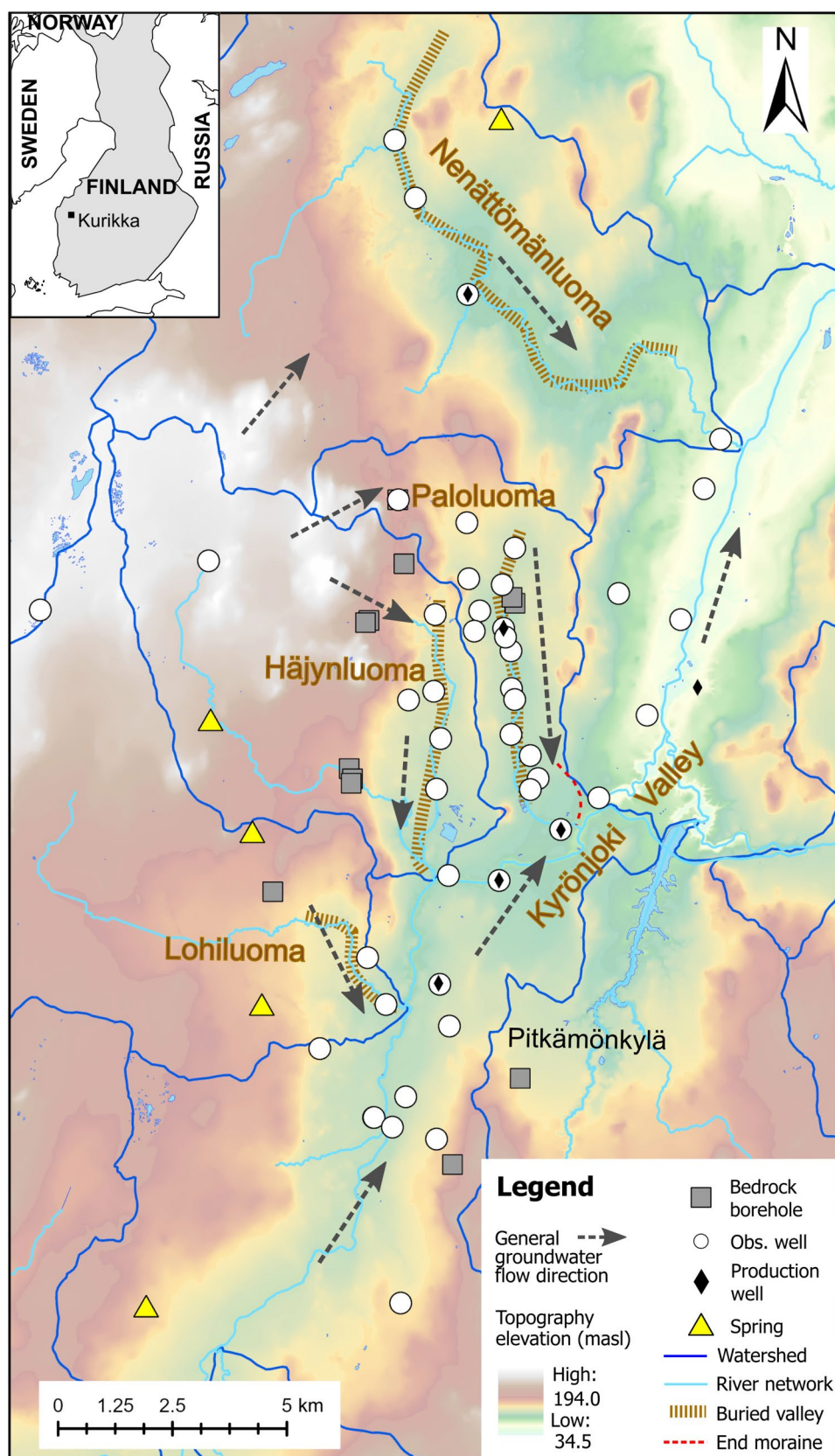
The study area (600 km²) is located in western Finland, near the municipality of Kurikka, within a subarctic climate zone. The landscape is a mix of forested and agricultural land. The topographic highs of the study area (194 m a.s.l.) are found in the west and the southeast (Pitkämönkylä area) while the valleys represent the lowest points, with elevations as low as 34.5 m a.s.l. The bedrock surface is below the modern sea level in many locations. The distance from the present shoreline is about 60 km. Four buried valleys are connected to the main Kyrönjoki valley located to the east of the study area: from north to south, Nenättömänluoma, Paloluoma, Häjyluoma, and Lohiluoma valleys (Fig. 1). In the study area, annual evapotranspiration is estimated to be in the range of 250–300 mm/year (Kolhinen et al. 2025) while the mean annual precipitation is 573 mm/year (Pirinen et al. 2022). Ongoing work indicates that average yearly recharge is in the order of 100 mm/year over the study area.

Geology and hydrogeology

Kurikka is located within the Archean and Proterozoic Fennoscandian Shield. The crystalline Paleoproterozoic bedrock in the study area is composed of paragneisses of turbiditic graywacke origin in the east, while the buried valley sediment aquifers and the western catchments are underlined by intrusive granodioritic–granitic–dioritic lithologies (Kähkönen 2005; Nironen 2005, 2017). Distinct N–S, NE–SW, and E–W-trending deformation or fault zones characterize the topography and contributed to the formation of the valley system (Putkinen et al. 2026; Ruuska et al. 2023).

The ancient craton underlying the region has undergone a complex, multi-phase history of burial and erosion (Hall et al. 2021). The prolonged periods of surface cover over the basement, i.e., crystalline bedrock, in conjunction with the area's position within the stable cratonic region, have contributed to the preservation of an extensive and ancient bedrock valley system that traverses the region (Putkinen et al. 2026). The area underwent its most

Fig. 1 Map of the study area showing the four buried valleys connected to the main Kyrönjoki buried valley. The map also shows the types of sampling locations (bedrock boreholes, observation wells (“obs. well”) in sediments, production wells in sediments and springs) and the general groundwater flow directions based on a regional piezometric map



recent phase of exhumation approximately 40 million years ago, initiating the latest stage of surface denudation characterized by intense tropical weathering, which altered the upper portions of the bedrock. Over tens of millions of years, sedimentary rocks that once covered the region were almost entirely removed, culminating during the Pleistocene glaciations, which progressively exposed the Kyrönjoki River valley along with the adjacent Nenättömänluoma, Paloluoma, Häjyluoma, and Lohiluoma sub-valleys (Fig. 1). This geomorphological evolution defined the major topographic features of the region, characterized by alternating hills and valleys. Meanwhile, the sedimentary infill at the base of the valleys was primarily deposited during the last glaciations, resulting in the present-day hydrogeological and sedimentological settings (Putkinen et al. 2026).

Åberg et al. (2026) described a highly complex sedimentary cover forming the unique aquifer system of the region. Moderate topographic variation, in combination with the heterogeneous sedimentary deposits and fractured bedrock, control the groundwater flow pathways. Infiltrated groundwater in the upland areas is transmitted through these routes and flows within the valley aquifers. Of particular significance is the lowermost coarse-grained sedimentary aquifer unit, which has demonstrated its suitability for high-rate municipal drinking water supply. This lowest sedimentary aquifer unit is confined beneath a sedimentary sequence that is tens of meters thick, composed of alternating aquifer and aquitard units. Together, these sedimentary units as well as the underlying fractured bedrock constitute a distinctive aquifer system, whose characteristics vary locally due to spatial differences in geological structure and composition across the study area.

Numerous hydraulic tests conducted in both the sediments and fractured bedrock indicated the presence of extensive, “highly” transmissive zones within the sediments, the bedrock, and their interface (Ruskeeniemi et al. 2023, 2024). Measurements and related investigations suggested that along the flow paths at the valley margins, particularly in sloping bedrock areas, there are multiple zones where the hydraulic conductivity of the local till, underlying bedrock and bedrock fracture zones and their interface is on the order of 10^{-6} m/s. Further downslope, toward the valley floors, groundwater becomes confined beneath aquitard layers composed of fine-grained lacustrine sediments and till. These aquitards overlie aquifer units whose hydraulic conductivities range from 10^{-5} to 10^{-4} m/s. In these confined zones, flowing artesian conditions are often present. Regionally, groundwater both discharges and flows predominantly eastward in the Kyrönjoki River valley region (Fig. 1).

The Nenättömänluoma Valley and the northern part of the main Kyrönjoki Valley are different from the other

valleys in the study area with more shallow sediment accumulations and poorly permeable conditions, with the exception of last deglaciation eskers occurrence at the top of the sedimentary sequence. These areas coincide with the principal groundwater recharge zones but are separated by the buried till-dominated end moraine zone (Fig. 1). The natural dam, formed from fine-grained tills within the buried end moraine and the surrounding fine-grained sediments, allows only slow seepage through it (Putkinen et al. 2026; Fig. 1), and together with the direction of groundwater flow from south to north, sets the groundwater level approximately 22.5 m higher than in the northern area. This creates a unique groundwater environment in the area, which has the greatest significance for public water supply and also exhibits potential for groundwater extraction; consequently, these northern valleys have been incorporated into the study.

The bedrock is exposed on scattered hilly areas. Sparsely fractured bedrock has been resistant to weathering and is therefore standing out in the topography. However, most of the bedrock is hidden under thick overburden in lowlands and can be studied only by drilling or geophysics.

Detailed investigations revealed a fracture network in the upper part of the bedrock occasionally extending as deep as to 120 m (Putkinen et al. 2026). A large part of the fracturing is formed within an extensional stress field, generating tensional fracturing with transmissivity. The fracture surfaces are rusted, indicating the circulation of fresh, oxic groundwater. Evidently, the recharging meteoric water has been weakly acidic, and the flow must have been active for considerable time because fracture-infilling calcites are depleted from the fracture network. Calcite reappears at the deeper levels where the other structural and mineralogical flow indicators are absent.

Structural analysis showed that subhorizontal to gently dipping fractures are common in the hydrogeologically active bedrock unit, consistent with the extensional stress and unloading concepts. Hydraulic testing confirmed that the horizontal hydraulic conductivity is consistently elevated (10^{-7} – 10^{-5} m/s) in the upper bedrock underlying the sedimentary aquifers situated in the valleys and in the area extending toward the western catchment. Bedrock boreholes in the center of the buried sedimentary aquifer units show flowing artesian behavior, thus indicating the groundwater from the bedrock is laterally connected to uplands and has a tendency to flow upward under valleys. Nearby boreholes are hydraulically connected, and the high yields observed in all the ten boreholes drilled in bedrock within the granular aquifer system suggest that the potential flow system within the bedrock is extensive. Four other boreholes beyond the southeastern water divide belong to a separate hydrogeological system.

The crust in the Kurikka region has repeatedly subsided into a depression during the Pleistocene glaciations. During these periods, and especially during deglaciation, fast-flowing glaciers further increased the depth of the Kurikka Valley, as the sediments filling the valley indicate multiple glaciations, deglaciation melting phases, and long-duration subaqueous sedimentation together with brackish-water intrusions (Putkinen et al. 2026). During the last deglaciation and the Holocene, the Kurikka area did not experience significant input of saline water, because the Baltic Sea water was more likely brackish. The peak salinity of the Littorina Sea has been estimated to be about 12 g/l (Westman et al. 1999). However, during the Eemian interglacial period (~135,000 years ago), Atlantic and Barents seawater spread to the Baltic Sea basin and these waters naturally extended as far as Kurikka. Subsequently, during the Weichselian Glaciation, those saline waters deposited muds that became mixed into the fine-grained regional till known as “Mäkisavi till” (Bouchard et al. 1990). These sediments are in contact with the Kurikka aquifer units and may contribute trace amounts of sodium, chloride, and metals to the groundwater. In addition, residual old marine waters may influence bedrock groundwater chemistry.

Data collection and analysis

The dataset used in this study is publicly available on Zenodo (cf. Availability of Data and Material). The corresponding sampling location names are mapped in Figure S1 of the electronic supplementary material (ESM).

Well construction and integrity of sampled wells

Drilling was conducted using casing tubes with an outside diameter (OD) of 101 mm and an internal diameter (ID) of 78 mm. Sediment samples were collected from the principal sedimentary units, and after the installation of the 60 mm OD and 50 mm ID groundwater observation well tubing, the steel casing tubes were removed. No sand packs or bentonite seals were used, as screens were installed directly in the primary deep aquifer unit.

Experimental field tests using active-distributed temperature sensing (A-DTS) fibers indicated that potentially vertical water flows across the observation well tube wall between the aquitards above and below the screened interval do not occur and do not compromise the aquifer investigations (Putkinen et al. 2026). This is due to the rapid collapse and compaction of the surrounding sediments against the well tubing shortly after installation, which effectively limits horizontal flow.

Groundwater sampling in sediments

All data described below were compiled into a single database including 160 samples across the study area. The methods used for sampling, chemical analyses, and data interpretation are described in the following sections.

Physicochemical data from previous phases of the Kurikka research project (2011–2021) were gathered from private well sampling and two bedrock boreholes, and included analyses of major and minor ions, trace elements, and field parameters. The analyses were carried out by different commercial laboratories (Eurofins Labtium and Botnia Lab/KVYY). The two bedrock boreholes sampled in 2021 were flowing under artesian conditions and samples were collected using a submersible pump to facilitate controlled sampling. These historical data were complemented with a regional geochemical sampling program conducted between June and September 2023. The 2023 sampling program focused on the unconsolidated sedimentary aquifers and two bedrock boreholes. A total of 53 water samples were collected from 34 observation wells in sedimentary aquifer units, two bedrock boreholes, six production wells in the sedimentary aquifer units and five springs. Observation wells and production wells in sediments have screened intervals between 2 and 65 m long (average interval length = 13 m). Groundwater samples were collected with an electrical submersible pump and spring water was collected as grab samples. In boreholes, at least three well volumes were pumped before sampling to ensure that the groundwater samples were representative of the sedimentary aquifers.

Field parameters including temperature, pH, electrical conductivity (EC), and dissolved oxygen (DO) were measured in the field using multiparameter probes (YSI ProDSS, or YSI/EXO Torpedo). Laboratory analyses for the 2023 campaign were conducted by a commercial laboratory (Eurofins Seinäjoki, Finland) for major ions (Ca^{2+} , Mg^{2+} , Na^+ , K^+ , Cl^- , SO_4^{2-} and NO_3^-) and minor and trace elements (Ag, Al, As, B, Ba, Be, Bi, Cd, Co, Cr, Cu, I, Fe, Hg, K, Li, Mn, Mo, Ni, P, Pb, Rb, S, Sb, Se, Sn, Si, SiO_2 , Sr, Th, Tl, U, V, Zn). Alkalinity was measured in the laboratory, sometimes a few days after sampling. This parameter may vary with time and is thought to be the cause of poor ionic balance for some of the analyses.

Isotopic analyses were conducted by the research laboratory of the Geological Survey of Finland (Espoo). Isotopic ratios of $\delta^2\text{H}$ and $\delta^{18}\text{O}$ were analyzed with a Picarro L2130i CRDS (cavity ring-down spectroscopy) analyzer. Isotope ratios of oxygen and hydrogen are reported using delta (δ) notation as measured $^{18}\text{O}/^{16}\text{O}$ or $^2\text{H}/^1\text{H}$ difference from the international VSMOW-standard as a per mil (‰). With L2130i the analytical uncertainty for $\delta^{18}\text{O}$ is <0.1 ‰ and for $\delta^2\text{H}$ <0.3 ‰. Strontium isotopic analyses (^{87}Sr , ^{86}Sr) were conducted by the same laboratory. After pre-treatment of

the samples, analyses were done with a MC-ICP-MS (multicollector-inductively coupled plasma–mass spectrometer) from Nu Instruments™ for samples collected between 2021 and 2022 and with a Neoma™ MS/MS, Thermo Scientific™ for samples collected in 2024. Measurement accuracy was monitored using the NBS987 standard.

Most locations were also sampled for ^3H analysis using 500 ml HDPE bottles or radiocarbon dating (^{14}C and $\delta^{13}\text{C}$) using 1L HDPE bottles. Seventy-six samples were selected for ^3H analysis and 49 were selected for radiocarbon dating. Analyses were done at the laboratory EIL Waterloo (University of Waterloo, Ontario, Canada) for ^3H and the Radiocarbon AMS facility of the University of Georgia for carbon-14. The selection of samples for isotopic analysis was conducted with the aim of maximizing spatial coverage and specifically targeting sediments and bedrock in different buried valleys, as well as springs to ensure good representativeness of each regional hydrogeological context.

Groundwater sampling in bedrock aquifers

Four additional groundwater sampling campaigns specifically targeted bedrock boreholes. Groundwater samples were collected from two boreholes in the Paloluoma valley in October 2020 and from six bedrock boreholes in the western part of the study area in November 2022. In April and October 2024, groundwater samples were collected from bedrock boreholes located in the western (eight boreholes) and southeastern parts of the study area (Pitkämönkyliä, four boreholes). Bedrock groundwaters were sampled using a tube sampler, allowing for the collection of a continuous water profile from the borehole (Nurmi and Kukkonen 1986). Depending on the length of boreholes, three to five groundwater subsamples were retrieved, each representing 20 to 25-m-long sections. The sampler is composed of a polyamide tube (diameter of 12/10 mm) with a back-pressure valve at the lower end and shutoff valves every 20–25 m. The sampler is slowly lowered into the borehole allowing groundwater to enter the tube. Once the bottom of the borehole is reached, the tube is lifted, the shutoff valves are closed, and the joints are separated. Water from the subsamples is retrieved for processing in field by nitrogen gas.

The same chemical parameters as previously described were analyzed, with the addition of bromide (Br). The October 2024 campaign did not include isotopic analyses and samples at two locations (KAL1 and KAL2) were only analyzed for ^3H . Temperatures from bedrock boreholes sampled in April 2024 were not all recorded in the field due to freezing conditions and a temperature of 5 °C was assigned to all missing values, based on the measurements of previous sampling campaigns and downhole electrical conductivity–temperature–depth (CTD) profiling. Additionally, sodium fluorescein dye added to the

drilling water was measured for the April 2024 samples. All boreholes were found to be representative for hydrogeochemical investigations as they did not include drilling water, except for two locations in the southeast (SAN1 and VAS1, not pictured).

Data processing

Several parameters of interest for data interpretation were calculated using the Diagrammes software (Simler 2012) or in Microsoft Excel: partial pressure of carbon dioxide (pCO_2), saturation indices, dissolved inorganic carbon (DIC), and ionic balance. All values reported as 'non-detected' (below the analytical method's detection limit) were replaced with a value equal to 50% of the detection limit (Sanford et al. 1993). The two bedrock boreholes that contained some drilling waters (see [Groundwater sampling in bedrock aquifers](#) section) were excluded from the analysis. The ionic balance acceptance limit was fixed at $\pm 15\%$ to maximize the number of analyses available for interpretation (Beaudry et al. 2018). Samples whose ionic balance is beyond this acceptable margin of error of $\pm 15\%$ are considered inadequate and were rejected. A total of seven samples had to be discarded for geochemical interpretation. Nevertheless, all the analyses for the bedrock boreholes were considered for the interpretation, even though they have a greater ionic balance and uncertainty due to their low mineralization.

Figure 1 shows the spatial distribution of the 67 sampling locations. Most of the observation wells and two production wells in sediments are located in the Paloluoma Valley. The bedrock boreholes are in the topographic highs of the study area in the Lohiluoma and Häjynluoma valleys, which are considered the recharge areas of the system (where bedrock outcrops or subcrops). Two other bedrock boreholes are located beneath the central Paloluoma Valley.

Groundwater types were first defined and represented on a Piper diagram which represents the proportion of major ions (Na^+ , Ca^{2+} , Mg^{2+} , K^+ , HCO_3^- , Cl^- , SO_4^{2-}) (Piper 1944). The Diagrammes software (Simler 2012) was used to create the Piper diagrams. Given the quantity of samples and the large number of chemical parameters, an analysis using statistical methods is necessary to process the data. Multivariate statistical analysis was successfully applied to understand the hydrogeochemical evolution of groundwater systems at the regional scale by numerous authors (e.g., Cloutier et al. 2006, 2008; Beaudry 2013; Montcoudiol et al. 2015; Beaudry et al. 2018).

Due to the large amount of geochemical data, the multivariate statistical analysis was combined with graphical hydrogeochemical interpretation. A detailed description of data preprocessing and multivariate statistical analysis is provided in the ESM.

The final dataset used for the multivariate statistical analysis resulted in 118 observations and 23 parameters. These parameters include pH, HCO_3^- , Si, Na^+ , SO_4^{2-} , Ca^{2+} , Cl^- , Mg^{2+} as well as minor and trace constituents F, Br, As, Fe, Mn, K, B, Ba, Cu, Mo, Ni, Sr, U, S, and Zn.

After selecting and standardizing the data, principal component analysis (PCA) and hierarchical ascending classification (HAC) were performed. Both PCA and HAC were performed using the package “FactoMineR” (Lê et al. 2008) and its hierarchical clustering on principal components (HCPC) function (Husson et al. 2010). PCA was used to identify the variables that explain most of the variation in chemical composition. In addition, unsupervised hierarchical clustering was applied to the PCA to group the samples according to their chemical similarity.

Corrected radiocarbon ages were calculated using the interpretation spreadsheet created by Janos (2017), which was also applied in Beaudry et al. (2018). This method identifies the geochemical processes that influence radiocarbon and selects the most appropriate correction methods for each sample.

Results

HCPC results

The descriptive statistics for the 118 groundwater samples selected for the HCPC are shown in Table S1 of the ESM. Table 1 presents correlations (loadings) between each variable and the principal components. The first two components together express 44.4% of the total dataset variance (25.3% for Component 1 and 19.1% for Component 2). The third component also accounts for 15.5% of variance. The first five components extracted have eigenvalues greater than 1, and account for 74.8% of the total variance in the dataset. Only the first two components will be further described as they were the ones used for the geochemical interpretation. The first component (Dim.1) is strongly positively correlated with Sr, HCO_3^- , Ca, Na, Mg, F, and K. The second component (Dim.2) is positively correlated with pH, U, Cu, Zn, and Ni, and negatively correlated with Si and Mn.

The hierarchical clustering was performed on the first five components of the PCA, and resulted in the definition of six clusters, or water groups, that are represented on the hierarchical tree (Fig. 2a). While the majority of the samples in clusters 1, 3, and 4 come from observation wells in sedimentary aquifer units, the majority of the samples in clusters 2, 5 and 6 come from bedrock boreholes. Observation of the hierarchical tree reveals some indications of the level of similarity between the six clusters. There is a statistical proximity between clusters 1 and 2, as well as between clusters 3 and 4. Clusters 5 and 6 are closely related and appear more

Table 1 Correlations between each geochemical variable and the principal components and their explained variance

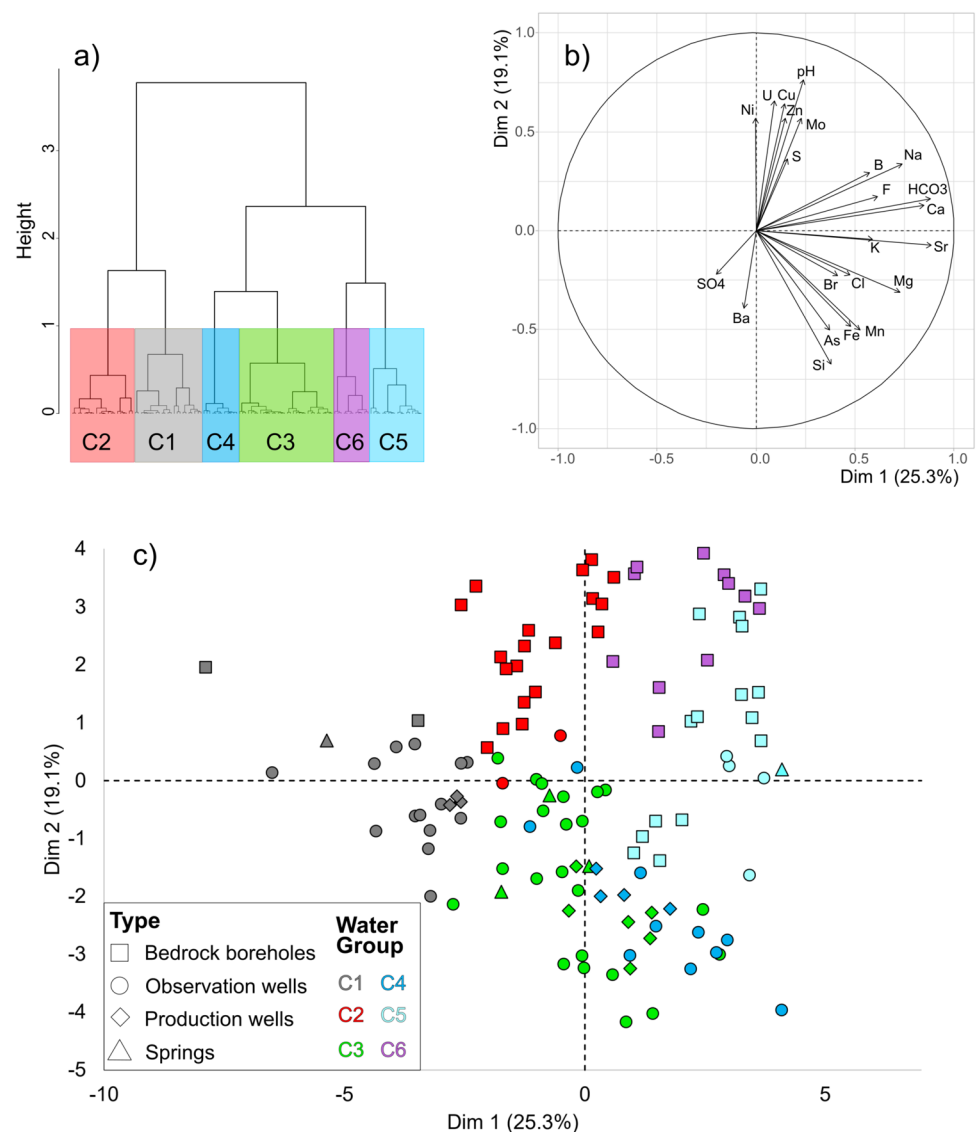
Parameter	Principal component				
	Dim.1	Dim.2	Dim.3	Dim.4	Dim.5
pH	0.241	0.763	-0.111	-0.280	-0.151
HCO_3^-	0.882	0.163	-0.196	-0.214	-0.085
F	0.615	0.172	-0.239	-0.356	0.148
Cl	0.474	-0.224	-0.430	0.541	0.226
Br	0.413	-0.226	-0.613	0.254	0.238
SO_4	-0.200	-0.220	0.211	0.798	-0.112
As	0.372	-0.501	0.405	-0.186	0.264
B	0.573	0.294	-0.504	-0.001	0.066
Ba	-0.060	-0.391	0.509	-0.029	-0.558
Ca	0.850	0.129	0.121	0.165	-0.352
Cu	0.144	0.641	0.529	0.019	0.254
Fe	0.479	-0.483	0.460	-0.123	0.258
K	0.590	-0.042	0.468	0.179	-0.058
Mg	0.728	-0.310	0.246	0.064	-0.374
Mn	0.524	-0.502	0.442	-0.150	0.121
Mo	0.227	0.567	0.419	-0.262	-0.090
Na	0.738	0.339	-0.423	0.115	0.136
Ni	-0.002	0.568	0.523	0.127	0.271
S	0.160	0.362	0.382	0.740	0.125
Si	0.380	-0.675	0.178	-0.173	0.273
Sr	0.885	-0.075	-0.074	0.108	-0.243
U	0.093	0.656	-0.095	0.115	-0.214
Zn	0.149	0.570	0.618	0.007	0.246
Eigenvalue	5.8	4.4	3.6	2.1	1.3
Percentage of variance (%)	25.3	19.1	15.5	9.0	5.8
Cumulative % of variance	25.3	44.4	59.9	69.0	74.8

distinct than the other clusters. Figure 2b shows the variable factor map in the plane defined by the first two components. The 118 samples are represented on the factor map produced by the first two principal components and colored according to their cluster in Fig. 2c. The first component is related to salinity because of the association of the variables Na, Cl and HCO_3^- . The second component is an important indicator of geochemical evolution through different parameters. It opposes groundwater samples from the bedrock (high pH, low Si, Fe, Mn, and As) to those from the sediments (lower pH, higher Si, Fe, Mn, and As) (Fig. 2b). Variables factor maps for the third and fourth components are presented in Figs. S2 and S3 of the ESM.

Spatial distribution of the water groups

Cluster 1 (C1) waters are present in all parts of the Häjynluoma Valley (Fig. 3) which corresponds to a buried

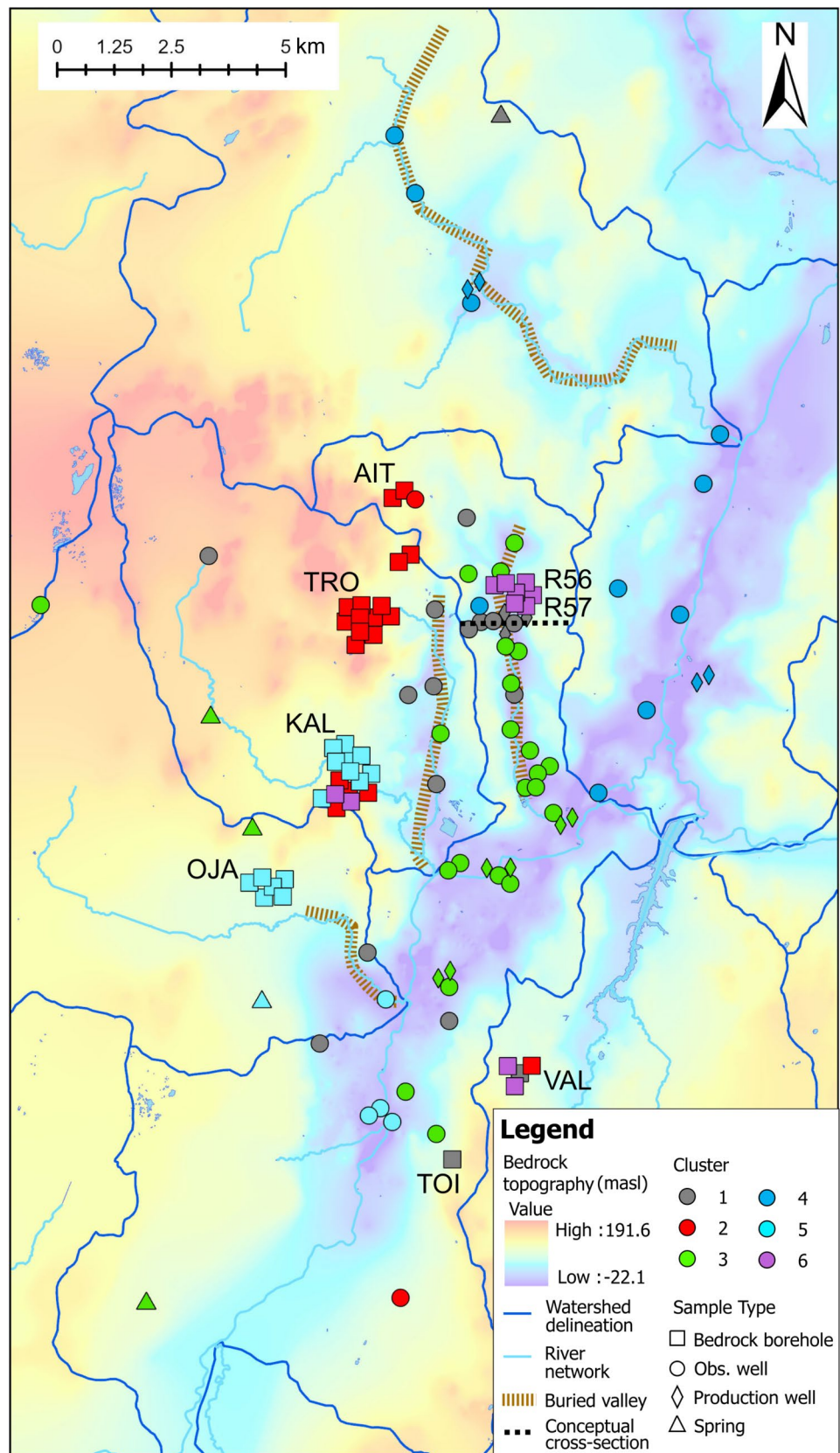
Fig. 2 Results of the HCPC **a** cluster analysis tree plot; **b** variables factor map; **c** Individuals factor map for the first two principal components of the analysis. The color code and shapes of the markers are applied in all subsequent figures. Factor maps of the third and fourth dimensions are provided in the ESM



esker. This less evolved groundwater type flows from north to south towards the main Kyrönjoki Valley and the predominance of C1 waters reflects the continuity of the groundwater flowpath within the buried esker. In the Paloluoma buried valley, the water type in observation wells in sediments changes from C1 and C3 in the upgradient area, to predominantly C3 in the central and downgradient areas. This may be because the hydrosystem becomes increasingly layered downgradient. C2 waters are found in bedrock groundwater samples and two observation wells in sediments in the topographic highs, in the west and southeast of the study area (Fig. 3). C4 waters are unique to the north and northeast of the study area, in the main Kyrönjoki and Nenättömänluoma valleys. This region is different from others in the study area as it is linked to a relatively modern esker and the aquifer system is shallower and not layered (see the *Data collection*

and analysis section). C5 waters are found in the south and southwest of the study area, both in samples from observation wells in sediments or springs and in bedrock groundwater (OJA and KAL bedrock boreholes). Geochemically evolved groundwaters (C6) are found in the bedrock deep beneath the sediments in the central Paloluoma Valley, as well as in the Pitkämönkylä area and in the southwest (KAL bedrock borehole). A depth profile analysis in bedrock shows that, for a given bedrock borehole, groundwater remains characterized by the same cluster in the shallowest and the deepest depth intervals of the well, except for the location VAL1 (Pitkämönkylä area). At VAL1, the groundwater is described by C2 (April) or C1 (September) between 2 and 30 m in depth, then changed to C6 between 52 and 76.8 m in depth. A more detailed analysis of the depth profiles in bedrock, including ³H analysis, is provided in the ESM.

Fig. 3 Location of the six clusters (water groups) defined by the statistical analyses. In some cases, multiple values are associated with the same coordinates (e.g., depth profiles in bedrock boreholes), and the symbols are therefore dispersed on the map for clarity. See Fig. S4 of the ESM for the details of the depth profiles in bedrock boreholes. “Obs. well” in the legend refers to observation wells



Hydrochemistry of the six water groups

Table 2 provides an overview of the median geochemical composition of clusters. This composition is also illustrated by Stiff

diagrams in Fig. 4a and the relative concentration of major ions is shown by the Piper diagram of Fig. 4b. C1 groundwater is characterized by the lowest concentration of many parameters, the lowest EC and an acidic pH of 6.5. C2 groundwater

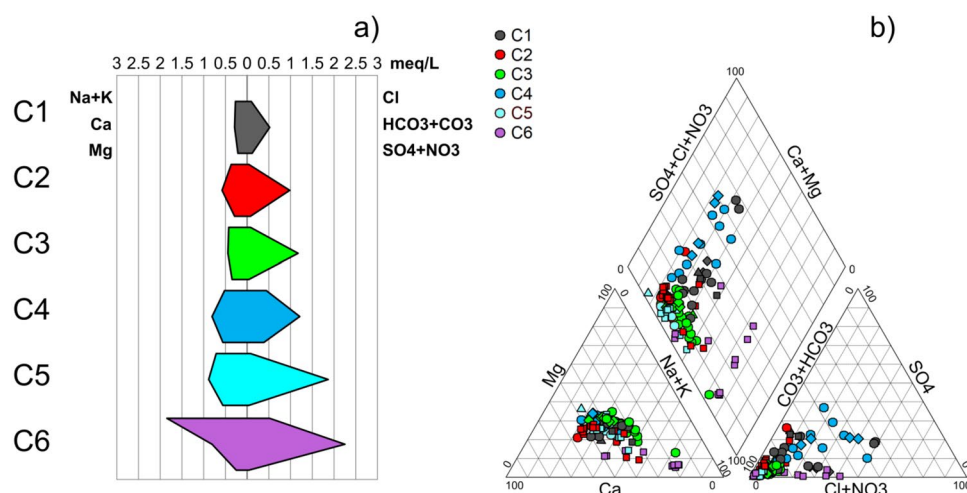
Table 2 Median water chemistry parameters by cluster. Concentrations are in mg/l, except for parameters marked with an asterisk (*). EC is electrical conductivity ($\mu\text{S}/\text{cm}$)

Parameter	Cluster					
	C1 <i>n</i> = 20	C2 <i>n</i> = 21	C3 <i>n</i> = 32	C4 <i>n</i> = 14	C5 <i>n</i> = 20	C6 <i>n</i> = 11
HCO ₃	34.2 ^b	61.0	70.1	66.2	114.9	135.5 ^a
Cl	1.9	1.4 ^b	2.9	15.5 ^a	2.6	14.0
SO ₄	2.8	3.3	1.9	19.0 ^a	1.8 ^b	2.3
Ca	6.2 ^b	11.0	8.1	16.5	18.0 ^a	13.0
Mg	2.7	3.8	4.0	6.4	6.6 ^a	2.6 ^b
Na	5.6 ^b	6.3	8.9	9.8	14.0	41.0 ^a
K	1.2 ^b	1.7	1.7	2.1	2.2 ^a	1.4
S	1.1	3.4	0.7 ^b	6.5 ^a	2.3	4.7
Si	11.9	8.8 ^b	14.3	14.5 ^a	14.0	9.4
Fe*	25 ^b	270	1525	3910 ^a	3250	84
Mn*	9.3 ^b	120	365	635 ^a	290	44
Sr*	45.2 ^b	62.0	69.0	98.0	115.0 ^a	92.0
Ba*	11.0	27.0	18.0	28.0 ^a	24.0	1.9 ^b
B*	3.8 ^b	7.3	7.8	7.4	13.5	30.0 ^a
Zn*	2.1	16	1.4	1.3 ^b	41.5 ^a	7.0
Ni*	0.64 ^b	5.0	0.56	1.3	2.8 ^a	2.0
Mo*	0.28	3.6	0.81	0.32 ^b	4.2 ^a	0.78
Cu*	0.42	3.1	0.21 ^b	0.37	5.0 ^a	2.1
As*	0.095 ^b	0.220	1.47 ^a	1.47 ^a	0.90	0.100 ^b
Br*	0.0079	0.0064 ^b	0.050	0.048	0.021	0.140 ^a
U*	0.03 ^b	0.14	0.03 ^b	0.04	0.07	0.25 ^a
pH	6.5	7.6	6.6	6.2 ^b	7.0	8.3 ^a
Depth (0–169 m)	58	98	62.5	34 ^b	65.5	114 ^a
EC	79.8 ^b	119	129.9	207.4	210.5	284.8 ^a

^aHighest values

^bLowest values

Fig. 4 **a** Stiff diagrams representing the average ionic compositions for each cluster; **b** Piper diagram representing the proportion of major ions in groundwater and spring samples, according to their respective cluster (water groups C1 to C6). Color codes and marker shapes as in Fig. 2c



is similar to C1 but relatively more mineralized and distinct by its slightly alkaline pH of 7.6. C1 and C2 are statistically close (Fig. 2a) and are respectively found mostly in sediments and bedrock, but their low concentrations and Ca–HCO₃ (or Ca–Mg–HCO₃) water types are typical of recently recharged groundwater.

Groundwaters of clusters C3 and C4 are statistically related (Fig. 2a) and represent more concentrated and evolved groundwaters found in sediments within buried valleys. The relatively high concentrations of Fe, Mn and As in these waters may reflect more reducing conditions. Similarly, groundwaters of clusters C5 and C6 are even more concentrated and reflect evolved groundwaters mostly found in bedrock. The neutral pH of C5 (7.0) and the alkaline pH of C6 (8.3) distinguish them from the acidic pH of C3 (6.6) and C4 (6.2).

Elevated levels of manganese and iron were generally found in all water groups (average concentrations were 350 and 2796 µg/l, respectively), both exceeding the drinking water guideline values in Finland (50 and 200 µg/l, respectively).

Major cations and anions were used to define the groundwater type. The proportions of major ions are presented on a Piper diagram (Piper 1944; Fig. 4b). The main anion is HCO₃ in all the samples, except two observation wells in the Paloluoma Valley for which chloride (Cl) is the dominant anion.

Most groundwater samples from the sediments and bedrock boreholes in the upgradient areas (C1, C2, C3, C5) are of Ca–HCO₃ or Ca–Mg–HCO₃ water types, reflecting less evolved groundwaters, typical of recharge areas (Cloutier et al. 2008; Beaudry et al. 2018). In contrast, groundwater from bedrock boreholes deep beneath the buried valleys (C6) showed a Na–HCO₃ type Na–HCO₃–Cl, reflecting more evolved groundwater. The stiff diagram (Fig. 4a) shows the distinct composition of each water type, based on average ionic composition for each cluster. These compositions reflect a geochemical evolution of groundwater from recharge areas to the buried valleys in the bedrock and sediments. The interpretation of this geochemical evolution relies on the spatial distribution of clusters (previous section) and on the geochemistry of the clusters. A similar groundwater evolution from Ca–HCO₃ to Na–HCO₃ water types was observed in both sediments and bedrock in the recharge area (C2–C5) (Fig. 4b). The line connecting water compositions in the cation triangle starts parallel to constant Mg and then curves down toward the Na apex, which is typical when ion exchange occurs (Hounslow 1995). Both Ca and Mg are progressively removed from the water and replaced by Na. However, except for the bedrock boreholes in C6, almost all samples are undersaturated with respect to calcite, which is due to the general calcite depletion in the

system (Putkinen et al. 2026). This overall lack of calcite likely indicates an active groundwater flow system over a very long-term span.

EC ranged from 27–382 µS/cm with a progressive increase from C1 to C6. Cluster 6 showed the highest EC values (> 200 µS/cm). Nevertheless, the overall EC values of the sampled waters are generally low, indicating low levels of mineralization. One bedrock borehole in C1 (TOI, Fig. 3) displayed very low EC values (27.1 µS/cm) up to a depth of 110 m, implying deep active groundwater recharge.

The partial pressure of CO₂ (pCO₂) provides insight on the degree of system “openness” with respect to soil CO₂. As water infiltrates the subsurface, it initially passes through the unsaturated soil zone, where it equilibrates with CO₂ produced by root respiration and the decomposition of organic matter. This results in pCO₂ levels that are higher than those in the atmosphere (Freeze and Cherry 1979; Appelo and Postma 2004). In open system conditions, water is in equilibrium with the soil zone CO₂ and pCO₂ remains constant as carbonate minerals dissolve until equilibrium is reached. Conversely, in a closed system where there is no subsurface source of CO₂ available, pCO₂ will gradually decrease and pH will increase.

The pCO₂–pH relationship (Fig. 5) shows a progression from open to closed system conditions across different areas. Sediment waters described by C1–C3–C4 reflect open system conditions, as well as bedrock waters from C5. In contrast, closed system conditions are found in bedrock groundwaters from C2 (Lehtivuori area, AIT1), and C6 (Paloluoma Valley and the Pitkämäonkylä area). This implies that radiocarbon dating can only be applied to the few bedrock boreholes samples under closed conditions.

An indication of the geochemical evolution of groundwater within the system is given by the relationship between pH and well depth (Fig. 6). There is a general trend in which pH increases with well depth. This illustrates the groundwater evolution from shallow, acidic waters from the sediments (C1–C3–C4) to deeper and more alkaline waters from the bedrock (C2–C5–C6).

Redox conditions can be indicated by Fe, Mn, and U levels in groundwater. Clusters 3, 4, and 5 showed the highest Fe and Mn concentrations and the lowest U concentrations, indicating reducing conditions. Conversely, C2 and C6 had relatively lower Fe and Mn concentrations and higher U concentrations, suggesting oxidizing conditions (Fig. 7, Table 2).

The strontium isotopic ratios range from 0.72 to 0.77, with a median value of 0.73, which is identical to that reported for regional estimates (Kaislaniemi 2011). The ratios display wide range in both sediment and bedrock samples, although sediment clusters 1 and 3 tend to show the highest values (Fig. S5 of the ESM). The isotopic composition of strontium

Fig. 5 Relationship between pH and $p\text{CO}_2$ values. Color codes and marker shapes as in Fig. 2c

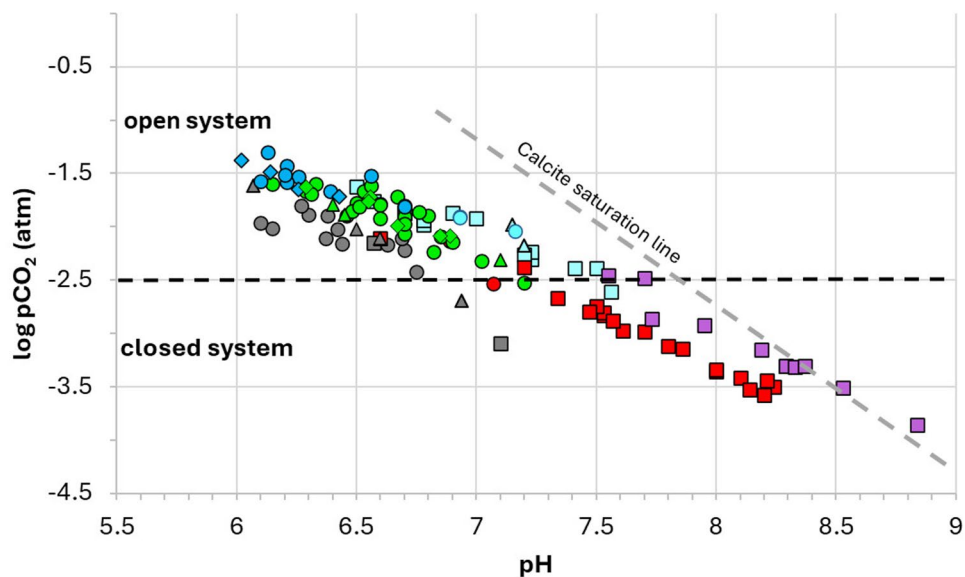


Fig. 6 Relationship between the pH of the groundwater samples and well depth (springs have a reported depth of 0 m). Color codes and marker shapes as in Fig. 2c

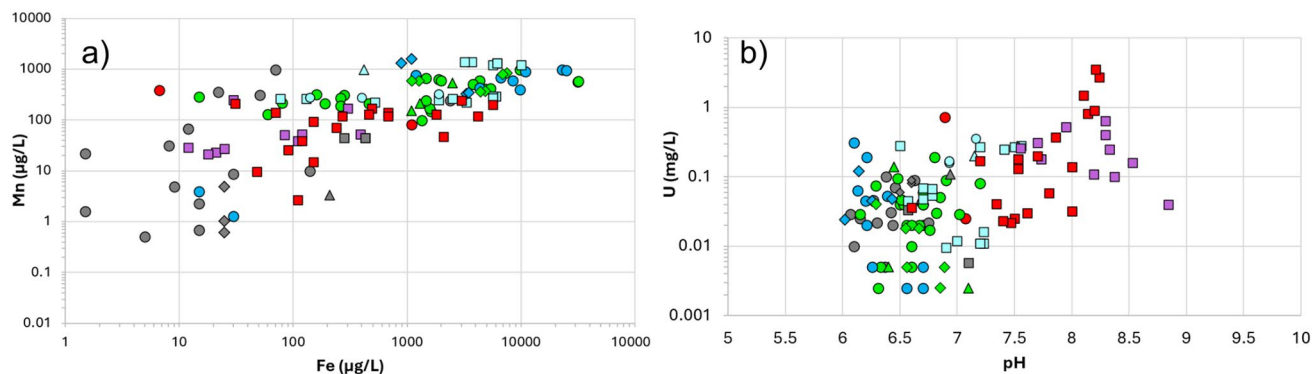
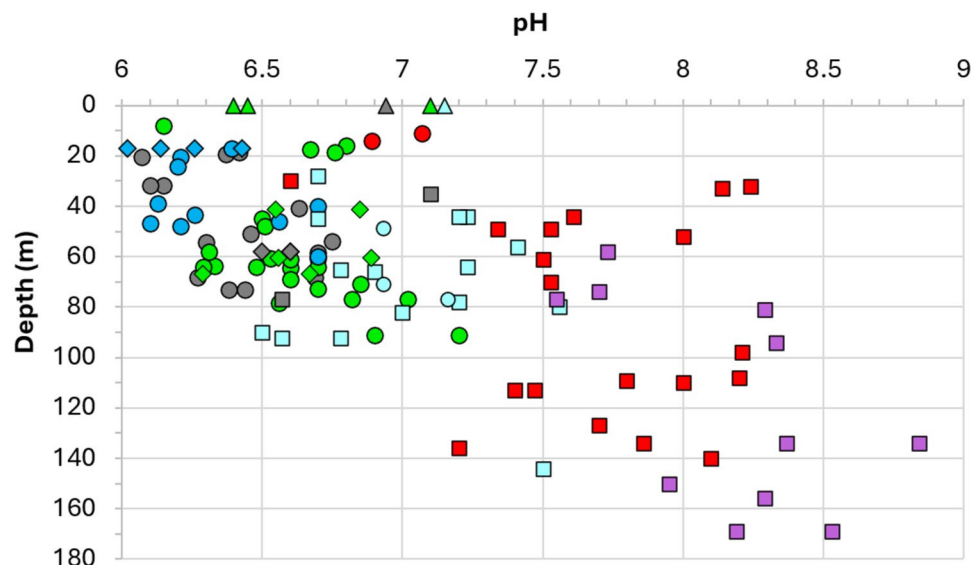


Fig. 7 Relationship between redox-sensitive elements and geochemical parameters: **a** Fe and Mn concentrations; **b** pH and U concentrations. Color codes and marker shapes as in Fig. 2c

reflects the geology of the study area, which is rich in gneiss, granite, and diorite with relatively high rubidium content. Strontium concentrations increase progressively from clusters C1, C2, and C3 to the other clusters, consistent with the geochemical evolution inferred from the other tracers in this study (Fig. S5 of the ESM).

The water stable isotope results ($\delta^{18}\text{O}$ and $\delta^2\text{H}$) show that all bedrock and sediment water samples plot along the Local Meteoric Water Line (LMWL) within a narrow isotopic range (1.2‰ for $\delta^{18}\text{O}$ and 10‰ for $\delta^2\text{H}$). This indicates that all water types are of meteoric origin and have not undergone significant evaporation or isotopic exchange (Fig. S6 of the ESM) and supports the interpretation of a hydraulically dynamic system.

Age dating

Tritium (^3H)

The closest GNIP station to the study site is Kuopio where ^3H in precipitation records are only available from 2005 onwards. Since the monthly ^3H values for Kuopio are very similar to the annual mean ^3H values in Vienna (Fig. 8), Vienna's long-term ^3H precipitation records were used as a proxy for this analysis. Therefore, the very long Vienna input curve (blue curve) was used as well as its decay-corrected curve scaled to 2021 (black curve). The decay-corrected curve represents the concentrations in ^3H that would be observed today for rainwater that fell in previous years. Using historical ^3H data from precipitation, the time period when the observed ^3H concentration in precipitation would have decayed to the measured concentration in groundwater in 2021 (year of sampling) was determined.

This provides an indication of the period of recharge, assuming no dispersion and no mixing with more recent or older water.

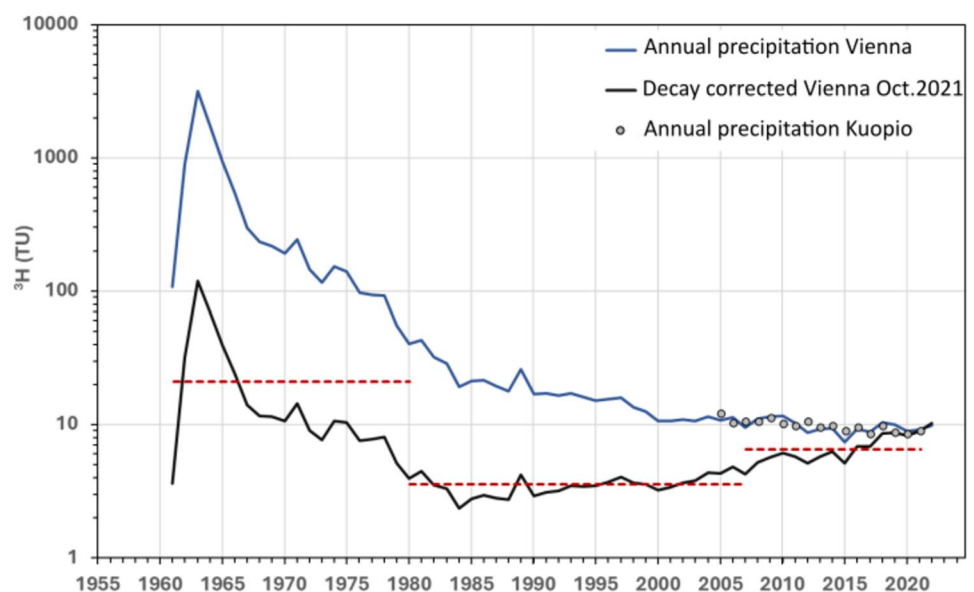
Recent water (< 15 years) has undergone limited effects of dispersion and mixing, so values near or above 5 TU are clear indications of a recent water. Mixing and dispersion can have more effect on older waters, so the mere presence (detection) of ^3H , even under 2 TU is an indication of at least a modern component as part of the water. Pre-1980 ^3H concentrations are rarely observed as these older waters have generally undergone much mixing with older (^3H -free) waters and dispersion, which reduce concentrations. Beyond 50 or 60 years, ^3H is not generally observed in groundwater originating from the infiltration of precipitation. The concentration of ^3H is thus a qualitative indicator of the occurrence of recent water (> 5 TU) or of modern component (< 50–60 years) in the water when ^3H is detected.

Thus, ^3H concentrations > 5 TU indicate a recent component (< 15 years) with low dilution. ^3H concentrations between 2 and 4.5 TU indicate a modern component (15 to 40 years). ^3H concentrations below 2 TU suggest the presence of a modern component with significant dilution with older water. The absence of ^3H (< 0.8 TU) indicates sub-modern waters with ages exceeding 60 years (and the absence of a modern component in this “old” water). ^3H concentrations over 6 TU could also indicate water between 40 and 60 years without mixing.

A modern component is present in C1, C2 and C4 waters, with ^3H concentrations generally between 2 and 5 TU. C3, C5 and C6 waters are generally ^3H -free and represent sub-modern waters.

Insight into the hydrosystem dynamics is provided by examining the relationship between ^3H concentrations and

Fig. 8 Average annual concentrations of tritium (^3H) in precipitation at Vienna (blue curve) and Kuopio (grey points), calculated from available IAEA monthly data. Average annual ^3H concentrations in groundwater calculated considering its degradation and scaled to 2021 (black curve). The red dotted lines correspond to average values assigned according to period



well depth (Fig. 9). At depths below 60 m, groundwater in the sediments is generally free of ^3H , suggesting a transit time of deep groundwater in buried valleys exceeding 60 years. In contrast, shallower zones (above 60 m) exhibit ^3H concentrations between 2 and 4 TU, indicating the presence of a modern water component and likely reflecting a local recharge to the sediments. While a few bedrock boreholes in the topographic highs and beneath the central Paloluoma buried valley showed a more evolved water type and were ^3H -free, most bedrock boreholes contained fresh and tritiated waters (from 2 to 9 TU) up to a depth of 135 m, suggesting deep active flow in bedrock. This indicates a very dynamic system with recharge groundwater reaching deep in the flow system. This could also be an indication of groundwater in bedrock reaching sediments.

A detailed analysis of the depth profiles in bedrock boreholes is presented in Fig. S4 of the ESM.

Springs were assigned to a depth of 0 m. Color codes and marker shapes as in Fig. 2c. Markers follow the color code and shape of the clusters and sample type, respectively. Uncolored markers correspond to bedrock boreholes samples that were not included in the statistical analysis, as they represent additional intervals in the depth profile.

Radiocarbon

Figure 10 relates the radiocarbon content in percent modern carbon (pmC) and the ^3H concentration, and it shows that most samples are “modern” as they plot above 75 pmC and contain ^3H . Therefore, the

Fig. 9 Relationship between well depth and tritium (^3H) concentration in groundwater samples

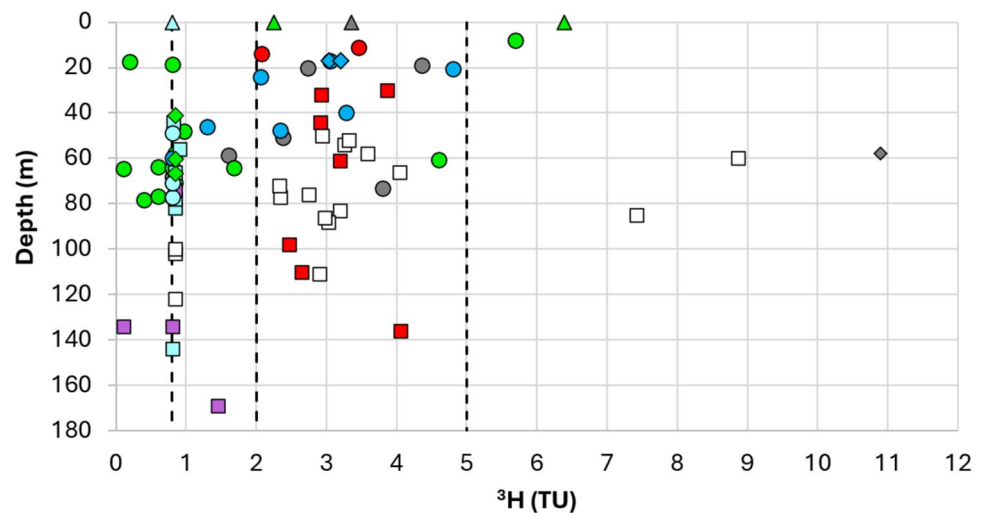
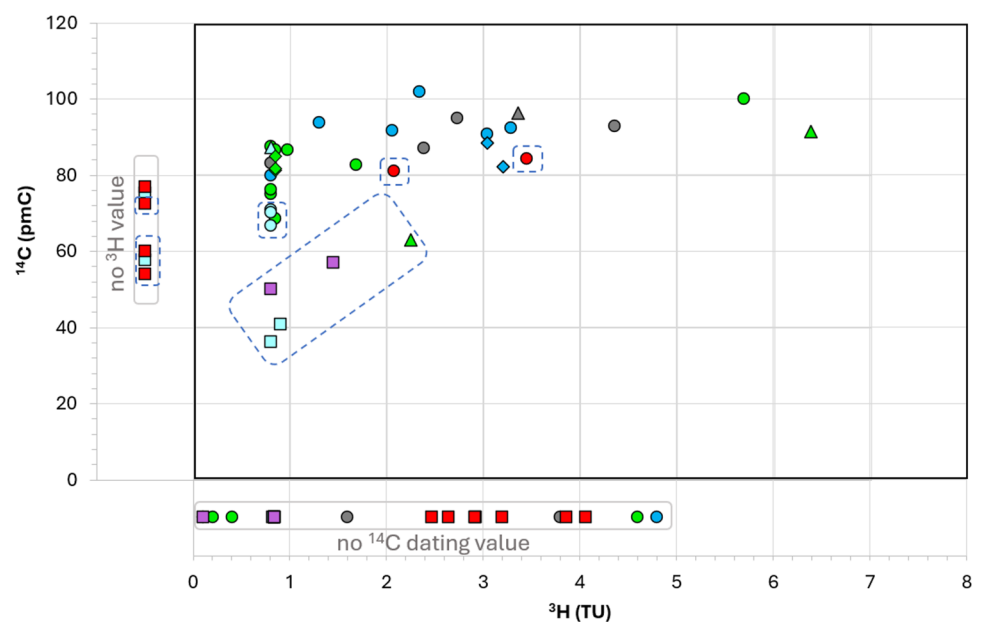


Fig. 10 Relationship between ^3H (TU) and ^{14}C (pmC). Circled markers indicate data points for which radiocarbon measurements provided information on relative groundwater age. Samples for which only one parameter was analyzed are represented in the left and bottom margin of the graph. Color codes and marker shapes as in Fig. 2c



radiocarbon measurements could only provide meaningful constraints for 12 samples, indicated in Fig. 10. In this context, radiocarbon results are not interpreted as absolute groundwater ages but rather as indicators of relative age relationships and of the presence of an “old water” component (> 60 years). In particular, when ^3H is low and the system is largely closed to CO_2 , corrected radiocarbon values corresponding to apparent ages greater than ~ 2000 years are interpreted as reflecting long residence times or mixing with an older groundwater component. The application of radiocarbon is limited in this system, as also evidenced by the large range of $\delta^{13}\text{C}$ values and the lack of trends reflecting processes such as calcite dissolution (Fig. S7).

The corrected radiocarbon ages were calculated using the interpretation spreadsheet proposed by Janos (2017). Table 3 shows the corrected ^{14}C residence time for the 12 samples for which the age determination was feasible, in addition to the laboratory (uncorrected) age and the simple corrected age (decay equation). In all cases, the Pearson model was the most suitable and the correction was small. Only five samples are clearly representative of old groundwaters, with residence times in the order of 4000 and above 7000 years. There is an indication that C5 and C6 have long residence times, as well as C2 in the deepest parts of the bedrock (TRO1 borehole).

Summary of hydrogeochemical and flow characteristics

Table 4 presents the main characteristics of the six clusters as well as their geochemical conditions and their implications for the groundwater flow, which are discussed in detail in the following Discussion section.

Discussion

Geochemical evolution

The identification of groundwater types with statistical analysis, along with the determination of residence times, provide a better understanding of regional groundwater flow and its geochemical evolution in the Kurikka bedrock and buried valley aquifer system. Table 4 summarizes the main characteristics of the six groundwater clusters and their residence times based on ^3H and radiocarbon. This table provides an overview of the evidence offered by groundwater clusters about the dynamics of the aquifer system.

Local complexities in the sediments and the bedrock as well as the layering influence the groundwater type. This is especially visible in the central Paloluoma Valley where three groundwater types are present. This occurrence of multiple types does not reflect mixing of groundwaters but rather the variety of origins of samples taken in shallow sediments, deep sediments and the underlying bedrock.

C1 waters are the least evolved and correspond to recharge waters in sediments. They are weakly mineralized, acidic and occur under open- CO_2 system conditions. Similarly, in the bedrock, C2 waters also correspond to recharge waters, as evidenced by tritiated waters observed at depths reaching 114 m. However, C2 waters showed higher pH values and low pCO_2 values associated with closed system conditions. This may reflect an evolution within bedrock, or a soil relatively depleted in organic matter, which would explain the drop in pCO_2 . The wide depth range of such modern groundwater could also imply active flow to great depths in the bedrock.

In the buried valleys, C3 waters represent a more evolved stage than C1. They are typically observed in sediments

Table 3 Summary of selected radiocarbon correction model, corrected age, uncorrected (laboratory) age, correction, sample's name, cluster and type of sample (sediments or bedrock)

Model	Corrected age (years)	Uncorrected lab age (years)	Simple age correction (years)	Correction (years)	Name	Cluster	Type
Pearson	1542	1330	1966	− 212	KARHU3	2	Sediments
Pearson	921	1660	1345	739	HARJA17	2	Sediments
Pearson	2305	2110	2729	− 195	AIT2-4	2	Bedrock
Pearson	1593	2580	2017	987	TRO1-1B	2	Bedrock
Pearson	3897	4110	4321	213	TRO1-4B	2	Bedrock
Pearson	2283	3670	2707	1387	LA58	3	Sediments
Pearson	2369	2270	2793	− 99	MIHP6b	3	Sediments
Pearson	5598	5520	6023	− 78	R56b	6	Bedrock
Pearson	4818	4470	5243	− 348	R57b	6	Bedrock
Pearson	1354	3230	1778	1876	LOHI24	5	Sediments
Pearson	6354	7180	6778	826	KAL2-1B	5	Bedrock
Pearson	7339	8150	7763	811	KAL2-5B	5	Bedrock

Table 4 Summary of cluster characteristics by quantitative variables, residence time indicators, geochemical conditions, and their implications on groundwater flow (GW) conditions

Cluster	Locations of samples and aquifer type	Depth ranges (m)	High values	Low values	Residence time (^3H range, TU)	Residence time (^{14}C corrected age)	Geochemical conditions (pH and redox)	Indication of flow conditions
C1	Paloluoma, Häjynluoma, and south of Kyrönjoki Valley (mostly granular aquifers), Pitkämäönkylä area (bedrock)	Wide range of depths (7–77 m)	None	HCO_3 , Ca, Sr, Fe, Mg, K, Mn, Na, F and As	Young to modern <0.8–10.9	n.a	Acidic (pH 6.5) and oxidizing	Young recharge GW in sediments
C2	Recharge areas in the west and northwest and the Pitkämäönkylä area (mostly bedrock boreholes)	Deep bedrock GW samples (30–140 m)	pH, Mo, Ni, Cu, Zn, U, S and Ba	Si, Br, Cl, Mn, Na and B	Modern 2.1–4.1	Modern, except for 1 location (3.9 kyr)	Alkaline (pH 7.6) and oxidizing	Young recharge GW in CO_2 closed system. High flow to depth
C3	Paloluoma Valley, south of the Kyrönjoki Valley and western part of the study area (wells in granular aquifers and springs)	Shallow to deep in sediments (0 to 90 m)	As, Si, Mn and Fe	S, Cu, Zn, Ni, U, Ca, pH and SO_4	Generally sub-modern <0.1–6.4	Modern	Acidic (pH 6.6) and a wide range of redox conditions	Moderately evolved GW in open CO_2 system
C4	Nenätömlänluoma, north of Kyrönjoki Valley (wells in granular aquifers)	Intermediate depths (17–60 m)	SO_4 , Cl, S, Mg, Ca, Sr, K, Fe, Mn and Br	pH, Mo, F and Zn	Modern <0.8–4.8	n.a	Acidic (pH 6.2) and reducing	Evolved GW but in open CO_2 system
C5	Recharge areas in the west and southwest (bedrock), Lohiluoma Valley and south of the Kyrönjoki Valley (granular aquifers)	Wide range relatively deep (44–140 m)	Mg, HCO_3 , Ca, Zn, K, Sr, Cu, Mo, Mn and F	SO_4	Sub-modern to modern <0.8–2.9	6.3–7.3 kyr	Neutral (pH 7.0) and highly reducing	Mixed and evolved GW in open CO_2 system
C6	Central Paloluoma Valley and the Pitkämäönkylä area (bedrock)	Deepest bedrock GW samples (76–156 m)	Na, B, pH, HCO_3 , F, Cl, Br, U, Sr and Ca	Ba, Fe, Mg, Mn, Si and As	Sub-modern <0.1–1.5	4.8–5.6 kyr	Alkaline (pH 8.3) and reducing	Very evolved GW in bedrock with CO_2 closed system

downgradient in the Paloluoma Valley or in the southern Kyrönjoki Valley over a wide range of depths (0 to 100 m). At greater depth in the bedrock or beneath the buried valleys, older and geochemically evolved waters of the C5 and C6 types are found. C4 waters are modern and rich in sulfates and sulfur and differ from other groundwaters in the rest of the system. They appear to be isolated from the rest of the water groups. Sulfide minerals are rare in the local crystalline bedrock and fractures, suggesting that the elevated sulfate concentrations are unlikely to originate from bedrock sources. Instead, sulfate may be derived from overlying Quaternary sediments. Large parts of the Kurikka area remained submerged for several millennia following deglaciation, during which thick sequences of lacustrine–marine fine sediments accumulated in topographic depressions (Silvennoinen et al. 2025). These deposits may represent a potential source of sulfate that is mobilized and flushed into the groundwater system, which could explain the distinctive chemistry of the C4 waters.

Overall, groundwater evolves from C1 or C2 to C3 in the sediments, and C2 to C5 and C6 in the bedrock, with mineralization and pH increasing as residence time grows.

The groundwater chemistry observed in the Kurikka buried valleys is broadly consistent with what has been reported in other buried valley aquifers, including studies from the United States (Rowe et al. 1999), Canada (Cloutier et al. 2006; Cummings et al. 2012), and Poland (Dragon 2006). In these systems, groundwater in recharge areas is typically characterized by Ca–HCO₃ or Ca–Mg–HCO₃ water types, with progressive geochemical evolution along the flow path toward waters enriched in Na, SO₄, and Cl. However, these studies report significantly higher total dissolved solids (TDS) concentrations than those observed in the Kurikka groundwater system, indicating comparatively lower mineralization in the study area. Several reference aquifers occur in sedimentary settings, which may partly explain their higher TDS concentrations compared to the crystalline bedrock environment of Kurikka.

The Na/Ca molar ratio (Fig. S8 of the ESM) illustrates the overall low sodium content in the hydrosystem and indicates that a slight effect of a freshening process (Na–Ca exchange) occurs in C5 and C6 waters. If there was a marine event in the system, some marine water would have been introduced. The fact that a marine “signal” is not observed (except in Harja 13, see the Database) suggests that almost all potential Cl[−] and Na⁺ were already flushed out of the system, along preferential flowpaths of high permeability both in the sediments and the bedrock. Thus, release of marine signal from adjacent sediments diminishes in time and becomes insignificant. Overall, the approach followed in this study proved effective for classifying the large number of groundwater samples and identifying the main hydrogeochemical processes controlling groundwater composition. This study

follows the recommendation of Güler et al. (2002) to combine multivariate statistical techniques with hydrogeochemical graphical methods in order to improve the interpretation of complex groundwater datasets. The multivariate statistical framework used here followed the methodology described by Cloutier et al. (2008), which has been widely applied in a variety of hydrogeological settings (Voutsis et al. 2015; Beaudry et al. 2018; Ren et al. 2021; Duan et al. 2022; Patel et al. 2023; Zhang et al. 2025; Sodomon et al. 2025) and is well adapted to the Kurikka hydrogeological context. These previous studies have generally used similar geochemical variable selections in multivariate analyses of hydrogeochemical datasets. Additional tracers, such as environmental isotopes or groundwater residence time indicators are also used to constrain recharge conditions and groundwater residence time, but they are not typically incorporated into the PCA as the method requires variables to be consistently available across the full dataset (e.g., Beaudry et al. 2018).

Implications for groundwater flow system

The geochemical evolution and spatial distribution of the water types have direct implications on the dynamics and connectivity between bedrock and sediment aquifer units of the groundwater flow system.

The general low mineralization in both sediment and bedrock groundwaters, even at depth, and the Ca–HCO₃ water type (except beneath the Paloluoma valley) indicate a largely dynamic flow system. Calcium concentration remains nearly constant across clusters, with only slight cation exchange effects in C5 and C6, consistent with a dynamic system or a lack of Ca²⁺ and Na⁺ sources.

This is also supported by ³H data, which indicates the presence of a modern component in most clusters (C1, C2, C4), except for some C3 sedimentary waters and bedrock groundwaters in C5 and C6. Thus, radiocarbon dating is of limited use in this system, as most samples exhibit open-system pCO₂ conditions, making ¹⁴C measurements unusable. In addition, δ¹³C values span a very wide range, preventing the use of simple graphical age-correction based on the relation between radiocarbon and δ¹³C (Han et al. 2012; Han and Plummer 2016). Only at limited locations (deep in the buried valleys or in the bedrock), is older water with a long residence time observed.

Consistency in groundwater evolution in the three smaller buried valleys (Paloluoma, Lohiluoma and Häjyluoma) indicates that they are part of a single regional groundwater flow system connected to the Kyrönjoki Valley. However, groundwater from the Nenättömanluoma and northern Kyrönjoki valleys (C4 water type) appears to be hydrogeologically disconnected from the southern part of the system as these regions are separated by the buried end moraine consisting

of fine-grained tills despite having the same recharge zone in the northwestern part of the study area.

In bedrock, the general presence of freshwater containing ^3H down to a depth as far as 110 m supports the concept of an active flow system in bedrock, recharged rapidly from the topographic highs of the study area. The regional topography provides an important driver for downward flow along a low-angle fracture network, while vertical fractures facilitate vertical connectivity attenuating with depth. Artesian conditions in the bedrock boreholes beneath the Paloluoma buried valley imply that fracturing is continuous from the higher locations of the study area to the lower part.

Flowing artesian conditions are common in the deepest aquifers of the buried valleys, due to the thick sequence of impermeable till and post-glacial and Holocene clay-silt and the connection of granular aquifer units to groundwater in topographic highs. Together with longer residence times, this indicates that production wells in southern Paloluoma and southern Kyrönjoki valleys, would be protected from anthropogenic effects, while still benefiting from groundwater renewal due to active recharge in connected upgradient topographic highs.

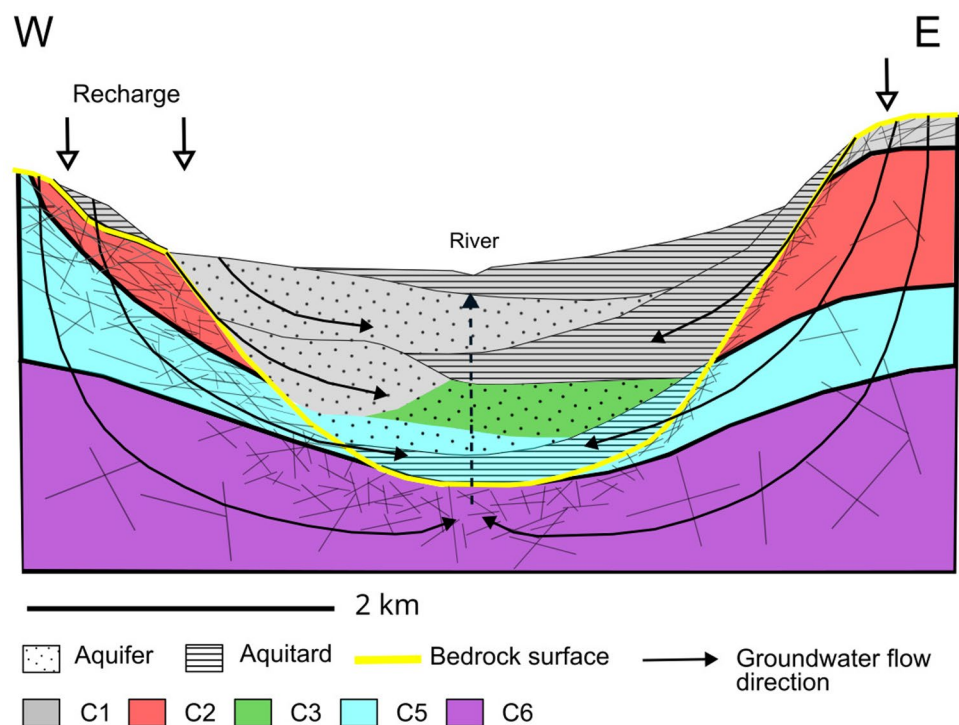
The conceptual cross section (Fig. 11) synthesizes the hydrochemical evidence into a simplified representation of groundwater flow and geochemical evolution in the sediments and upper fractured bedrock. This illustrates how water types evolve from modern recharge where bedrock outcrops or subcrops to geochemically evolved waters in deeper valley aquifer units and fractured bedrock. The colors in Fig. 11 are consistent with those used in the clustering.

Conclusions

A network of buried valleys with glaciofluvial sediments up to 120 m thick connected to fractured and transmissive bedrock has the potential to meet the growing demand for water in western Finland. A compilation of historical hydrochemical data, supplemented by a new regional sampling campaign in five buried valleys and the underlying fractured bedrock, was conducted to improve the understanding of the hydrosystem. Multivariate statistical analysis and residence time indicators showed that a modern water component and active flow system were present in all parts of the system. Longer residence times are found deep within the buried valleys (^3H -free waters) and at greater depth in a few locations in the bedrock (radiocarbon ages > 5000 years). Deep, downgradient areas in buried valleys with evolved water type indicate suitable water supply for production, protected from anthropogenic influences.

Combining hydrochemical data and multivariate statistical analysis is a transferable approach for characterizing complex aquifer systems. This study illustrates how the geochemical evolution of groundwater can be used as an indicator of flow conditions, geological history, and the mineralogical composition of aquifer units in sediments and bedrock. Long residence time in bedrock and in certain deep parts of the buried valleys demonstrates their relative isolation, which is essential for sustainable exploitation and the natural protection of the resource against contaminants (low vulnerability).

Fig. 11 Conceptual cross section of the geochemical evolution in the sediments and the fractured upper bedrock in the Kurikka buried valley system. The general location of the cross section is shown in Fig. 3. Water types (C1 to C5) are described in Table 4. Arrow thickness indicates the magnitude of groundwater flow. Dashed arrows indicate inferred groundwater flow. No vertical scale implied



More research is needed to better uncover the relationship between bedrock and the sediments. To achieve this, a sampling scheme including additional locations beneath the Paloluoma and Kyrönjoki valleys would be necessary. This study and the resulting conceptual understanding of the groundwater geochemical evolution will support the development of a numerical groundwater flow model of the system and will contribute to informing future management decisions. More broadly, the findings illustrate methodological and conceptual approaches relevant to groundwater management in buried valleys globally.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10040-026-03166-4>.

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Data availability The datasets generated and/or analyzed during the current study are available at Zenodo (<https://doi.org/10.5281/zenodo.16746379>). However, coordinates are not publicly available due to the nature of the commercial project and sensitivity of the ongoing water abstraction licensing. Access may be considered on a case-by-case basis upon reasonable request to Niko Putkinen niko.putkinen@gtk.fi.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no conflicts of interest.

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